Autoencoders for Image Reconstruction and Generation

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I. Introduction

The purpose of this investigation is to evaluate how different autoencoder architectures and latent space configurations affect reconstruction accuracy, generative capability, and information flow. Specifically, we compare a modified deterministic convolutional autoencoder with a convolutional variational autoencoder, examining the impact of Gaussian noise and varying latent dimensionality. Through qualitative reconstruction comparisons and quantitative measures of information rate, this study aims to highlight the trade-offs between latent representation efficiency, reconstruction fidelity, and the ability to generate meaningful outputs.

II. Theory

This section discusses the underlying principles used in the model creation.

A. Autoencoder

Autoencoders are neural networks designed to learn efficient representations of data. They consist of two main components: an encoder, which maps the input data into a lower-dimensional latent space, capturing the most important features, and a decoder, which reconstructs the original input from this latent representation.

a) Basic Autoencoder: A basic autoencoder encodes each input into a single latent vector, which the decoder uses directly to reconstruct the input. The model is trained by minimizing a reconstruction loss, such as mean squared error or binary cross-entropy, which measures the difference between the reconstructed output and the original input.

$$x \xrightarrow{f_\phi} z \xrightarrow{g_\theta} \hat{x} \tag{1}$$

b) Variational Autoencoder: A variational autoencoder encodes each input into two vectors representing the mean and log-variance of a Gaussian latent distribution. A latent vector is then sampled from this distribution and passed to the decoder to reconstruct the input. The model is trained by minimizing the Evidence Lower Bound (ELBO), which balances reconstruction accuracy and regularization of the latent space.

$$x \xrightarrow{f_\phi} \mathcal{N}(\mu, \sigma^2) \xrightarrow{z\sim(\mu,\sigma^2)} z \xrightarrow{g_\theta} \hat{x} \tag{2}$$

B. Loss Functions

a) Mean Squared Error (MSE): The mean squared error computes the average squared difference between the model's predicted and expected outputs.

$$MSE = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 \tag{3}$$

MSE loss is used to train basic autoencoders as it directly penalizes element-wise differences between the input and its reconstruction. This is equivalent to maximizing the likelihood that the reconstructed data matches the original under a Gaussian observation model.

b) Evidence Lower Bound (ELBO): The Evidence Lower Bound (ELBO) loss combines a reconstruction term and a regularization term. The reconstruction term measures how well the reconstructed data matches the original input, while the regularization term, given by the Kullback–Leibler divergence, encourages the latent distribution to align with a standard Gaussian prior.

$$\text{ELBO} = \mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x|z)\big] - D_{\text{KL}}\big(q_\phi(z|x)\|p(z)\big) \tag{4}$$

C. Gradient Descent

Gradient descent iteratively updates the parameters of an autoencoder to minimize its loss function, with backpropagation used to compute gradients.

D. Information Rate

The information rate quantifies the mutual information between the input data and the latent representation, indicating how many bits of information effectively pass through the network's bottleneck. Added Gaussian noise in the latent layer reduces the effective information rate, as some of the latent signal is obscured. Conversely, a higher latent variance relative to the noise allows more information to pass through, improving the model's capacity to reconstruct the input.

$$I(z; z_{\text{noisy}}) = \frac{1}{2}\sum_{i=1}^{d} \log_2\left(1 + \frac{\sigma_i^2}{\sigma_{\text{noise}}^2}\right) \tag{5}$$

III. Experiments

A. Setup

Two convolutional autoencoder neural network models were designed and trained. The models were compared based on their reconstruction accuracy and generation quality.

a) Input data: Each model receives a 28×28 grayscale image containing a single shape — circle, square, or triangle — with random size and position, and produces a reconstruction of that image as output.

$$x \in \mathbb{R}^{1\times 28\times 28} \quad \mapsto \quad \hat{x} \in \mathbb{R}^{1\times 28\times 28} \tag{6}$$

b) VAE Model: Convolutional Variational Autoencoder: The convolutional VAE consists of a two-layer convolutional encoder and a two-layer transposed convolutional decoder. The encoder maps the input to a latent distribution with a user-defined dimensionality, which can be adjusted to control the size of the latent space. The decoder reconstructs the input from a sampled latent vector.

c) AE Model: Convolutional Autoencoder with Gaussian Noise: This model is a modified basic convolutional autoencoder with a two-layer convolutional encoder and a two-layer transposed convolutional decoder. Gaussian noise is added to the latent layer during training, and the latent vector is regularized using variance control to encourage an approximately Gaussian distribution. The model is trained using Mean Squared Error (MSE) loss to minimize reconstruction error between the input and output.

d) Hyperparameters: All experiments were conducted over 20 epochs with a batch size of 64, a learning rate of 0.001, and the Adam optimizer. The Gaussian noise magnitude was set to a constant 0.3. The latent dimensionality varied based on the experiment.

B. Model Comparison

These experiments compare each model's reconstruction accuracy and its ability to generate images from Gaussian latent vectors, using a latent dimension of 16. Outputs are compared qualitatively, as quantitative comparison is not feasible due to differing loss functions.

a) Reconstruction: The modified AE produced more accurate reconstructions of the original images compared to the VAE (Fig. 1), despite the addition of Gaussian noise to the latent vector. This is likely because the noise is small and the latent representation remains close to the deterministic encoding, whereas the VAE's stochastic sampling can produce blurrier outputs.



Fig. 1: Reconstructed images from each model.

b) Generation: The VAE generates better images (Fig. 2) because its latent space is regularized to follow a Gaussian distribution, ensuring random samples decode into meaningful outputs. Whereas the modified AE's latent space is less structured and only optimized for reconstruction.

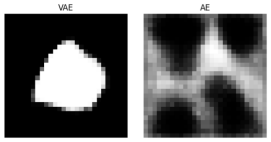


Fig. 2: Generated images from each model.

TABLE I: Information Rate in Bits

	2 dims	**8 dims**	**16 dims**	**32 dims**
0.1 noise level	6.02	7.32	7.11	6.43
0.5 noise level	3.73	5.08	4.71	4.08
1.0 noise level	2.92	4.05	3.77	3.11

C. Information Rate

Table 1 shows that increasing Gaussian noise in the latent layer reduces the information rate, while increasing the latent dimensionality initially raises it, peaking around 12 dimensions before declining. Higher noise obscures the latent signal, and beyond a certain dimensionality, additional latent dimensions contribute little new information relative to the noise, lowering the overall information rate.

D. Latent Space Dimensionality

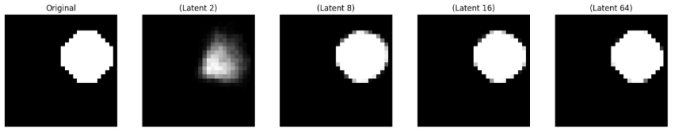


Fig. 3: Reconstructions using different latent dimensionalities.

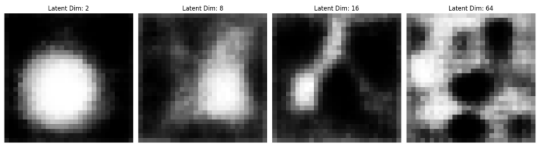


Fig. 4: Generated images using different latent dimensionalities.

The latent space dimensionality determines the size of the latent vector(s). Larger latent dimensions can capture more input information, but they also increase model capacity and may result in inefficient or redundant representations.

Figures 3 and 4 show four models trained with the modified AE, each using a different latent dimensionality. While higher dimensionality improves reconstruction accuracy, generation quality declines, as many latent samples fall in unused regions of the space and fail to capture meaningful information.

IV. Conclusion

The modified autoencoder produced more accurate reconstructions, while the variational autoencoder better structured the latent space for generation. Gaussian noise in the latent layer reduces the effective information rate, and increasing latent dimensionality initially improves information throughput but eventually adds little new information. Despite this, higher latent dimensions improved the visual quality of reconstructed images, highlighting a trade-off between information efficiency and perceptual fidelity.

STATEMENT

This report and the code provided were entirely my own work.

Tools used: Overleaf, Google Colab, Kaggle, PyTorch, PIL, NumPy, Matplotlib.

Source code is available from the following repository: https://github.com/CRM48/victoria_projects/tree/main/neural_networks/autoencoders.

It contains a ReadMe file which contains a link to the code.

It also contains the ipynb file which can be opened in Kaggle by clicking the link on the file.

The VAE model designs and training processes were inspired by Justin Barry [1] and Reza Kalantar [2].

REFERENCES

[1] J. Barry, "Building a Variational Autoencoder (VAE) from Scratch in PyTorch — MNIST with Linear Layers," YouTube, Oct. 6, 2023. [Online]. Available: https://www.youtube.com/watch?v=Nn20SUGY6qs

[2] R. Kalantar, "Variational Autoencoder (VAE) — PyTorch Tutorial," Medium, 20-Nov-2022. [Online]. Available: https://medium.com/@rekalantar/variational-auto-encoder-vae-pytorch-tutorial-dce2d2fe0f5f